GEOAI 04 change report: Bihar (Patna Ganga floodplain)

Totals

Raw detections: 2049
Real changes after filters: 79
Area changed: 201.52 ha
Cropland affected: 20.28 ha (upper bound)
Built-up land affected: 3.61 ha
critical: 2
high: 8
medium: 16
low: 53

Open this result: /scene/bihar-encroachment

Top 10 changes

#	Tier	Type	Area (ha)	Conf.	Reason
1	critical	water_loss	168.675	0.981	water receded: vegetation index rose (+0.35), water index fell (-0.46), SWIR/blu
2	critical	water_loss	2.59	0.985	water receded: built-up index rose (+0.21), vegetation index rose (+0.30), water
3	high	built_up_gain	2.885	0.962	new construction / built-up: vegetation index fell (-0.07), water index rose (+0
4	high	water_loss	2.535	0.967	water receded: built-up index rose (+0.22), vegetation index rose (+0.42), water
5	high	built_up_gain	2.085	0.967	new construction / built-up: water index rose (+0.06), SWIR/blue ratio (paved su
6	high	built_up_gain	2.36	0.957	new construction / built-up: vegetation index fell (-0.12), water index rose (+0
7	high	built_up_gain	1.945	0.948	new construction / built-up: SWIR/blue ratio (paved surfaces are flatter) fell (
8	high	built_up_gain	0.825	0.972	new construction / built-up: vegetation index fell (-0.14), water index rose (+0
9	high	built_up_gain	0.93	0.961	new construction / built-up: vegetation index fell (-0.09), water index rose (+0
10	high	built_up_gain	0.99	0.95	new construction / built-up: vegetation index fell (-0.08), water index rose (+0

Sources and limits

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0
- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Rivers, sandbars and lake edges move between dates and can be flagged as change; bare soil in dry regions can look like construction. Check flagged areas near water.
- Land cover is ESA WorldCover 2021, newer than the before image (2019): land already cleared or under construction can be mislabelled as cropland, so treat cropland as an upper bound
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero